Opening Questions

Sports

- Who is the best quarterback of all time?
- Are baseball pitchers throwing harder than ever?
- How much more do male soccer players earn than females?
- How common is it for former Olympic athletes to become coaches?
- How much does an extra inch of height help a basketball player?

Pop Culture

- What percentage of people have seen the movie that won last year's Best Picture Award?
- Who tends to be more popular: bands or solo singers?
- Are younger actors paid more than older actors?
- Are movies with female leads as profitable as movies with male leads?
- Does winning a Grammy increase sales?

Politics

- Is "Stop and Frisk" a racist policy?
- Do Republican politicians tend to come from different states than Democratic ones?
- Do people in countries that have universal healthcare live longer than people in countries that don't?
- Was press coverage slanted for or against a particular candidate?

Education

- Do small schools perform better than large ones?
- Which has a stronger correlation with student achievement: race or wealth?
- Do bilingual classes result in better outcomes for ESL/ELL students?

What Questions Can You Answer with the Given Data?

The following is a dataset of a bicycle rider's training rides.

date	miles	time (w/stops)	weather	average speed	max speed
04/10/2018	10	44	"cloudy"	13	30
05/30/2018	15	66	"sunny"	13.5	22
06/12/2018	12	61	"rainy"	11.2	25
07/04/2018	24	103	"sunny"	14	26
07/12/2018	24	120	"windy"	12.5	26

1) Decide whether each questions below *can* or *cannot* be answered with the given data and circle your selection.

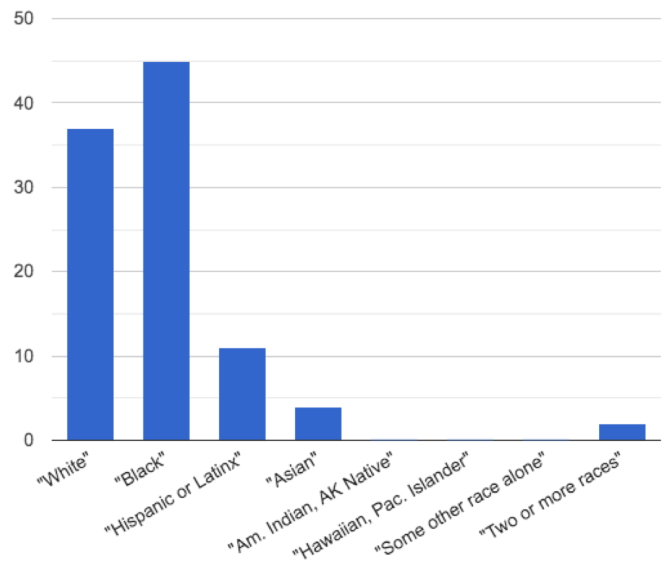
2) In the space provided below, explain *how* you could answer the question using the data or *why you cannot* answer the question.

Question	Answered by this dataset?
How many miles did the cyclist ride June 12th?	*YES* _No
What tire pressure produces the highest average speed?	Yes *NO*_
What is the average time it takes this cyclist to ride 1 mi?	*YES* _No
Does this cyclist ride slower when it is rainy?	Yes *NO*_
Does this cyclist ride faster when they are late to an appointment?	Yes *NO*_
How many miles has the cyclist ridden in total as part of their training?	*YES* _No

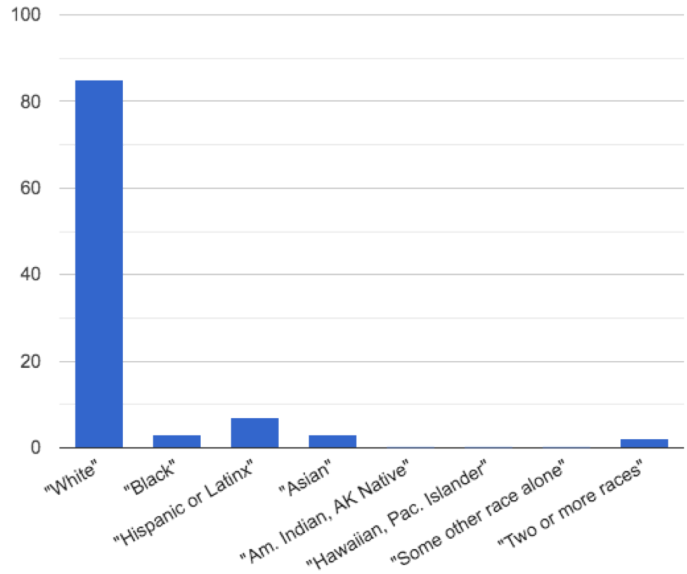
Are there any questions that you could find the answers to more than one way?

Bar Chart - Notice and Wonder

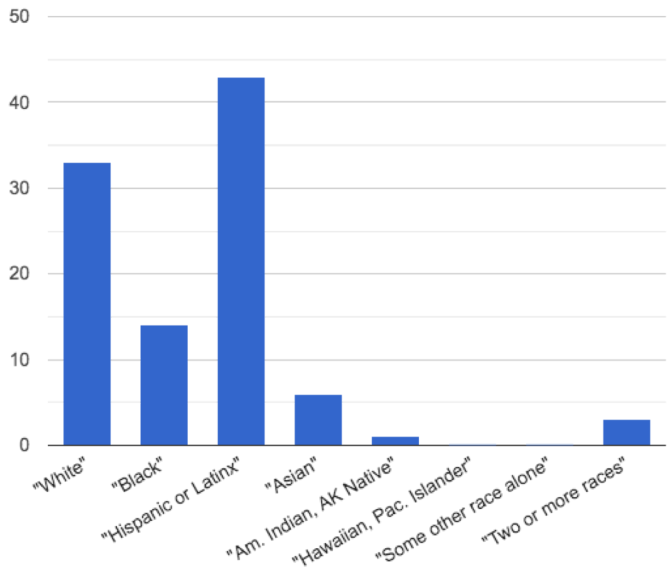
Washington, D.C. Public Schools



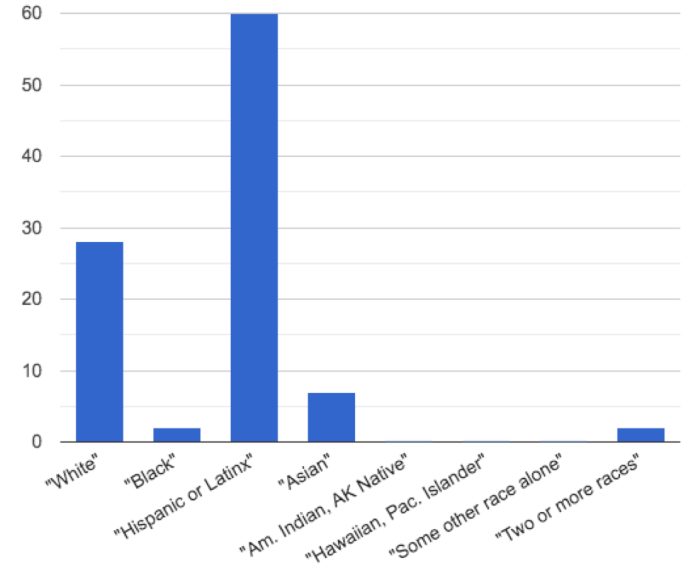
West Warwick Schools, RI



Providence Schools, RI



Oxnard Union HS District, CA



What do you Notice?	What do you Wonder?

Question Types: Animals

A subset of the whole Animals Dataset is shown in the table below.

name	species	sex	age	fixed	legs	pounds	weeks
Sasha	cat	female	1	false	4	6.5	3
Sunflower	cat	female	5	true	4	8.1	6
Felix	cat	male	16	true	4	9.2	5
Sheba	cat	female	7	true	4	8.4	6
Billie	snail	hermaphrodite	0.5	false	0	0.1	3
Snowcone	cat	female	2	true	4	6.5	5
Wade	cat	male	1	false	4	3.2	1
Hercules	cat	male	3	false	4	13.4	2
Toggle	dog	female	3	true	4	48	1

Using this table - or the full dataset - write three questions of each type below. ***Student answers will vary, but sample questions are provided below***

	Question	Type
1	*How old is Boo-Boo?*	Lookup
2	*How long did it take Toggle to get adopted?*	Lookup
3	*Is Fritz fixed?*	Lookup
4	*Which animal is the oldest?*	Arithmetic
5	*What is the average age of the animals?*	Arithmetic
6	*How many of each species are there?*	Arithmetic
7	*What is a typical adoption number of weeks to be adopted?*	Statistical
8	*Do heavier animals get adopted more slowly?*	Statistical
9	*Are dogs adopted faster than cats?*	Statistical

Expression Cards

Print and cut these out, for use with the `Filter` Transformer activity in this lesson. You can make up your own using the blanks provided at the bottom!

Do they wear glasses? `Student -> Boolean` consumes a student from the dataset, produces true if they are wearing glasses	**Are they taller than you?** `Student -> Boolean` consumes a student from the dataset, produces true if they are taller than you
Do they have long hair? `Student -> Boolean` consumes a student from the dataset, and produces true if they have long hair	**Do they have red hair?** `Student -> Boolean` consumes a student from the dataset, and produces true if they have red hair
Are they shorter than you? `Student -> Boolean` consumes a student from the dataset, and produces true if they are shorter than you	**Are they smiling?** `Student -> Boolean` consumes a student from the dataset, and produces true if they are smiling
Does their name begin with a vowel? `Student -> Boolean` consumes a student from the dataset, and produces true if the first letter of their name is a vowel	**Are they wearing jeans?** `Student -> Boolean` consumes a student from the dataset, and produces true if they are wearing jeans

What Table Do We Get?

The table below represents four animals from the shelter:

name	sex	age	fixed	species	pounds
"Toggle"	"female"	12	true	"dog"	48
"Fritz"	"male"	4	false	"dog"	92
"Nori"	"female"	6	true	"dog"	35.3
"Sunflower"	"female"	2	false	"cat"	51.6

Below is a list of saved Transformers. What table do we get, after applying each one? *Match* each expression*left_* to the table we get*_right*.

`order-by-age`
Transformer: Sort
Attribute to Sort By: age
Direction: ascending

`order-by-weight`
Transformer: Sort
Attribute to Sort By: pounds
Direction: descending

`filter-is-old`
Transformer: Filter
Formula to Filter By: Age > 10

`filter-is-fixed`
Transformer: Filter
Formula to Filter By: `Fixed = true`

`filter-is-dog`
Transformer: Filter
Formula to Filter By: `species = "dog"`

`build-is-dog`
Transformer: Build Attribute
Formula for New Attribute Values: `species = "dog"`

`build-is-fixed`
Transformer: Filter
Formula to Filter By: `fixed = true`

1 - *B*

2 - *F*

3 - *C*

4 - *E*

5 - *A*

6 - *D*

7 - *G*

A Produces a table with Toggle, Fritz, and Nori - but not Sunflower.

B Produces a table of all four animals, sorted youngest-to-oldest

C Produces a table, with only Toggle.

D Produces an identical table with an extra column called "dog", whose values are true, true, true, false

E Produces a table containing only Nori and Toggle.

F Produces a table with all four animals, sorted from heaviest to lightest.

G Won't run: will produce an error

Table Transformations with Chaining

You have created and saved the following Transformers. (Read them *carefully!*)

`filter-is-fixed`	`filter-is-young`	`filter-is-female`	`build-kilos`
Transformer: Filter Expression: Fixed = true	Transformer: Filter Expression: Age < 4	Transformer: Filter Expression: Sex = “female”	Transformer: Build Attribute Name of new attribute: Kilos Expression: Weight (lbs) / 2.2
`filter-is-heavy`	`filter-is-dog`	`filter-is-cat`	
Transformer: Filter Expression: Kilos > 25	Transformer: Filter Expression: Species = “dog”	Transformer: Filter Expression: Species = "cat"	

Solve the word problems below using the Transformers listed above.

	Prompt	Does order matter?	List the transformers in order.	Use function notation to predict the table’s name.
1	Create a table of all female cats.	No	Apply `filter-is-female` . Apply `filter-is-cat` .	filter-is-cat(filter-is-female(animals-table))
2	Create a table of all young, fixed animals.	*No*	*Apply* `filter-is-young` . *Apply* `filter-is-fixed` .	*filter-is-young(filter-is-fixed(animals-table))*
3	Create a table of all female dogs, converting their weight to kilograms.	*No*	*Apply* `filter-is-female` . *Apply* `filter-is-dog` . *Apply* `build-kilos` .	*filter-is-female(filter-is-dog(animals-table))*
4	Create a table of all heavy dogs.	*Yes*	*Apply* `filter-is-dog` . *Apply* `build-kilos` . *Apply* `filter-is-heavy` .	*filter-is-dog(build-kilos(filter-is-heavy(animals-table))*
5	Create a table of all young, fixed dogs, sorted from lightest to heaviest.	*No*	*Apply* `filter-is-young` . *Apply* `filter-is-fixed` . *Apply* `filter-is-dog` .	*filter-is-young(filter-is-fixed(filter-is-dog(animals-table)))*

Predictions from Samples

1) Use the Sampler plugin to create the following tables and give them the specified names. (Re-open Sampler *before* creating each table to avoid getting one massive hierarchical table!)

- 10 animals randomly selected without replacement ⇒ tiny-sample
- 20 animals randomly selected without replacement ⇒ small-sample
- 40 animals randomly selected without replacement ⇒ medium-sample
- 80 animals randomly selected without replacement ⇒ large-sample

2) Make a bar chart of the species in the tiny sample.

- What animals are in the sample? _____
- Create a new random tiny-sample. Create another bar-chart of species in the tiny-sample. What animals are in the sample? _____
- Create another random sample and make another bar-chart of species in the tiny-sample. Based on these samples, how many species of animals do you think are at the shelter? _____
- Which species do you think there are the most of at the shelter? _____

3) What did you learn from taking multiple samples that you wouldn't have known if you'd only taken a single sample?

_____

4) Now use the small sample to make a bar chart of the species.

- What animals are in the sample? _____
- Create a new random sample and make another bar chart of species in the small sample. What animals are in the sample? _____

5) Now that you've seen the small sample, how has your sense of the distribution of the species changed?

_____

6) Now use the medium sample to make a bar chart of the species. If there are about 400 animals at the shelter, how many of each species would you predict there to be?

_____

7) Now use the large to make a bar chart of the species. If there's anything you'd like to change about your prediction now that you've seen the large sample, record it here.

_____

8) Let's see how accurate your prediction is… When you're ready, make a bar chart of Animals Table 2.

- Which predictions were closest? _____
- Which predictions were off? _____
- Were there any surprises? _____

9) In the real world, we usually don't have access to a whole dataset to check predictions against! How could we test…

- *Every giraffe on the planet?*
- *Everyone who has ever come in contact with a covid-positive person?*
- *Every person who identifies as queer?*

What strategies can we use to make sure that predictions from samples are as close to accurate as possible?

_____

_____